



Regd. Office : Aakash Tower, 8, Pusa Road, New Delhi-110005, Ph.011-47623456

CONCEPT STRENGTHENING SHEET

CSS-05

Chemistry

ARBTS-14

Q.88. (Cannizzaro Reaction)

1. Cannizzaro reaction is not shown by

- (1) HCHO (2) $\begin{array}{c} \text{CHO} \\ | \\ \text{CHO} \\ | \\ \text{CHO} \end{array}$
- (3) $\text{CH}_3\text{CH}_2\text{CHO}$ (4) $\begin{array}{c} \text{CHO} \\ | \\ \text{C}_6\text{H}_5 \end{array}$

2. Among the following the rate of Cannizzaro reaction is slowest for

- (1) $\begin{array}{c} \text{CHO} \\ | \\ \text{C}_6\text{H}_5 \\ | \\ \text{OH} \\ | \\ \text{CHO} \end{array}$ (2) $\begin{array}{c} \text{CHO} \\ | \\ \text{C}_6\text{H}_5 \\ | \\ \text{SH} \\ | \\ \text{CHO} \end{array}$
- (3) $\begin{array}{c} \text{CHO} \\ | \\ \text{C}_6\text{H}_4 \\ | \\ \text{NH}_2 \end{array}$ (4) $\begin{array}{c} \text{CHO} \\ | \\ \text{C}_6\text{H}_5 \end{array}$

3. Among the following the rate of Cannizzaro reaction is maximum for

- (1) $\begin{array}{c} \text{CHO} \\ | \\ \text{C}_6\text{H}_4 \\ | \\ \text{CH}_3 \end{array}$ (2) $\begin{array}{c} \text{CHO} \\ | \\ \text{C}_6\text{H}_5 \end{array}$
- (3) $\begin{array}{c} \text{CHO} \\ | \\ \text{C}_6\text{H}_4 \\ | \\ \text{NO}_2 \end{array}$ (4) $\begin{array}{c} \text{CHO} \\ | \\ \text{C}_6\text{H}_4 \\ | \\ \text{OCH}_3 \end{array}$

4. $\text{CCl}_3\text{-CHO}$ can undergo

- (1) Aldol reaction only

(2) Cannizzaro reaction only

(3) Both Aldol and Cannizzaro reactions

(4) Neither aldol nor Cannizzaro reactions

5. $\begin{array}{c} \text{CHO} \\ | \\ \text{CHO} \end{array} \xrightarrow{\text{conc. KOH}} \text{A (Major)}$

Product A is

- (1) $\begin{array}{c} \text{CH}_2\text{OH} \\ | \\ \text{CH}_2\text{OH} \\ | \\ \text{COOK} \end{array}$ (2) $\begin{array}{c} \text{COOH} \\ | \\ \text{COOH} \\ | \\ \text{COOK} \end{array}$
- (3) $\begin{array}{c} \text{COOH} \\ | \\ \text{CH}_2\text{OH} \end{array}$ (4) $\begin{array}{c} \text{COOH} \\ | \\ \text{CH}_2\text{OK} \end{array}$

ARBTS-15

Q.51. (Periodic Table)

1. If value of ℓ can vary from 0 to n then maximum number of elements in 2nd period will be

- (1) 2 (2) 8
- (3) 18 (4) 32

2. Total number of elements theoretically possible in 8th period is

- (1) 50 (2) 48
- (3) 32 (4) 18

3. Element containing electrons in 4d orbital can not be present in

- (1) 4th period (2) 5th period
- (3) 6th period (4) 7th period

4. Which of the following orbitals is not filled in 5th period

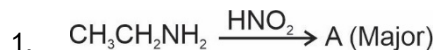
- (1) 5s (2) 4d
- (3) 5p (4) 4f

5. 5g orbitals filling takes place in

- (1) 5th period (2) 6th period
- (3) 7th period (4) 8th period

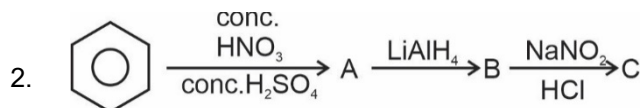
ARBTS-15

Q.82. (Diazotisation)

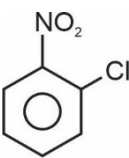


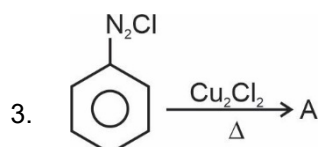
Product A is

- (1) $\text{CH}_3\text{CH}_2\text{NO}_2$ (2) CH_3NH_2
(3) $\text{CH}_3\text{CH}_2\text{OH}$ (4) CH_3OH

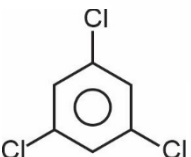
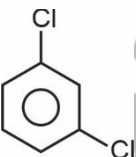


The product C is

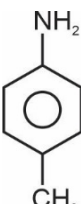
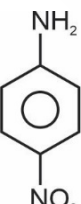
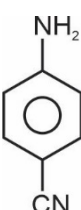
- (1)  (2) 
(3)  (4) 

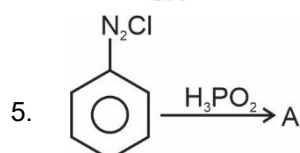


Product A is

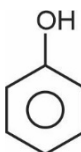
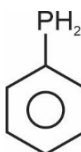
- (1)  (2) 
(3)  (4) 

4. Aryl amine which undergoes diazotization most easily is

- (1)  (2) 
(3)  (4) 



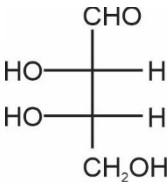
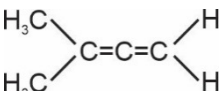
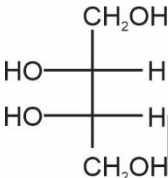
The product A is

- (1)  (2) 
(3)  (4) 

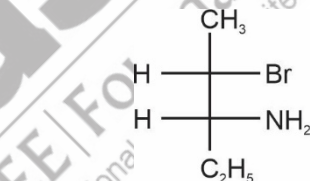
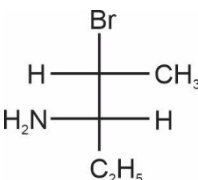
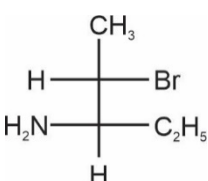
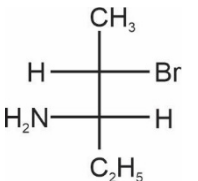
ARBTS-14

Q.80. (Optical isomerism)

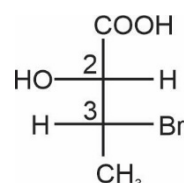
1. Which among the following is/are optically active compound(s)?

- (1)  (2) 
(3)  (4) Both (1) & (2)

2. Which among the following is the enantiomer of the given compound?

- (1)  (2) 
(3)  (4) 

3. Assign R, S configuration to 2 and 3 carbons respectively in the given compound



- (1) R, R (2) S, S
(3) R, S (4) S, R

4. For the molecule $\text{HOOCCH(OH)CH(OH)COOH}$ total number of stereoisomers and meso form respectively are

- (1) 4, 0 (2) 3, 1
(3) 2, 1 (4) 3, 2

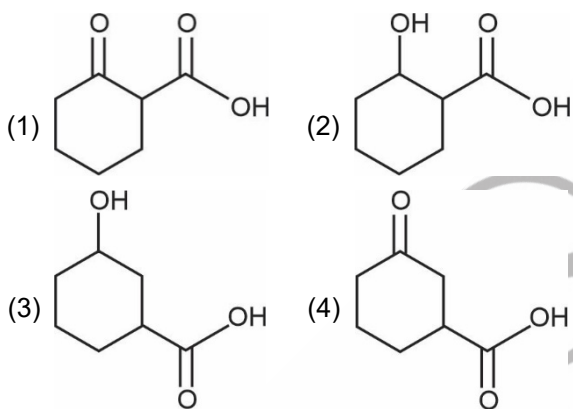
5. Total number of optically active stereoisomers for the compound $\text{CHOCH(OH)CH(OH)CH(OH)CH}_2\text{OH}$ will be

- (1) 4 (2) 8
(3) 6 (4) 10

ARBTS-14

Q.81. (Decarboxylation Reaction)

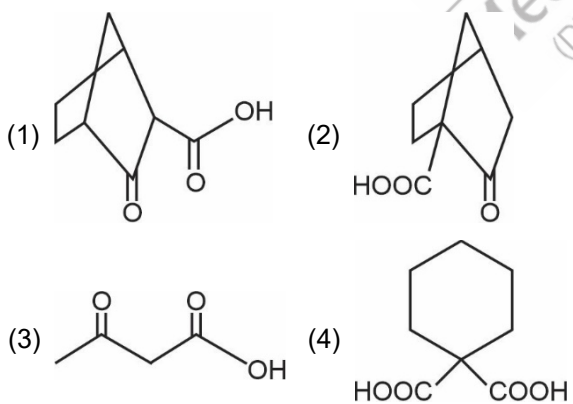
1. Which compound will decarboxylate most easily on heating?



2. When malonic acid is heated at $140-150^\circ\text{C}$ then the product formed is

- (1) HCOOH (2) CH_3COOH
(3) $\text{CH}_2=\text{CH}-\text{COOH}$ (4)

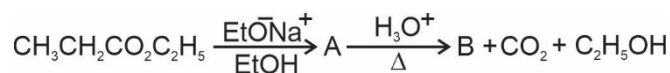
3. The compound which is most difficult to decarboxylate is



4. Pyruvic acid when treated with warm dilute sulphuric acid then the product formed is

- (1) $\text{CH}_3-\text{C}(=\text{O})-\text{CH}_3$ (2) $\text{CH}_3\text{CH}_2\text{OH}$
(3) CH_3CHO (4) HCHO

5. Consider the following reaction



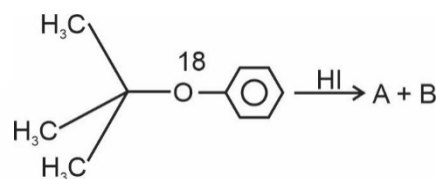
Product B is

- (1) $\text{CH}_3-\text{C}(=\text{O})-\text{CH}_3$ (2) $\text{CH}_3\text{CH}_2-\text{C}(=\text{O})-\text{CH}_2\text{CH}_3$
(3) $\text{CH}_3\text{CH}_2\text{CH}_2\text{COOH}$ (4) $\text{CH}_3\text{CH}_2\text{CH}_2-\text{C}(=\text{O})-\text{CH}_3$

ARBTS-14

Q.82. (Reaction of Ethers)

1. Consider the following reaction



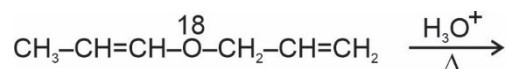
Products A and B are

- (1) $(\text{CH}_3)_3\text{C}-\text{OH}$ and
- (2) $(\text{CH}_3)_3\text{C}-\text{I}$ and
- (3) $(\text{CH}_3)_2\text{C}=\text{CH}_2$ and
- (4) $(\text{CH}_3)_3\text{C}-\text{I}$ and

2. The ether which is most difficult to cleave by heating with HI is

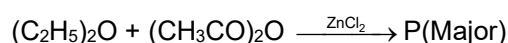
- (1) $\text{CH}_3\text{CH}_2-\text{O}-\text{CH}_3$
(2) $\text{Ph}-\text{CH}=\text{CH}-\text{O}-\text{CH}_3$
(3) $\text{Ph}-\text{CH}_2-\text{O}-\text{CH}_2-\text{Ph}$
(4) $\text{Ph}-\text{O}-\text{Ph}$

3. Major products formed in the given reaction are



- (1) $\text{CH}_3\text{CH}_2\text{CHO}$ and $\text{CH}_2=\text{CH}-\text{CH}_2-\text{OH}$
(2) $\text{CH}_3\text{CH}_2\text{CHO}$ and $\text{CH}_2=\text{CH}-\text{CH}_2-\text{OH}$
(3) $\text{CH}_3-\text{CH}=\text{CH}-\text{OH}$ and $\text{CH}_2=\text{CH}-\text{CH}_2-\text{OH}$
(4) $\text{CH}_3-\text{CH}=\text{CH}-\text{OH}$ and $\text{CH}_2=\text{CH}-\text{CH}_2-\text{OH}$

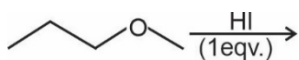
4. Consider the following reaction



Major product P is

- (1) $\text{CH}_3-\text{C}(=\text{O})-\text{OC}_2\text{H}_5$ (2) $\text{C}_2\text{H}_5-\text{C}(=\text{O})-\text{OCH}_3$
 (3) $\text{CH}_3-\text{C}(=\text{O})-\text{C}_2\text{H}_5$ (4) $\text{CH}_3-\text{C}(=\text{O})-\text{C}_3\text{H}_7$

5. Major products of the given reaction are

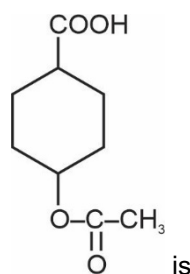


- (1) MeI and $\text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$
 (2) MeOH and $\text{CH}_3\text{CH}_2\text{CH}_2\text{I}$
 (3) MeOH and $\text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$
 (4) MeI and $\text{CH}_3\text{CH}_2\text{CH}_2\text{CHO}$

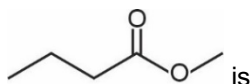
ARBTS-13

Q.57. (Nomenclature)

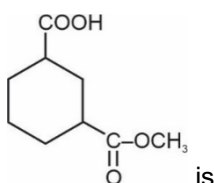
1. The IUPAC name of the compound



- (1) 4-carboxycyclohexane ethanoate
 (2) 4-ethanoylcyclohexane carboxylic acid
 (3) 4-(ethanoyloxy)cyclohexane carboxylic acid
 (4) 4-propanoylcyclohexane carboxylic acid
2. The IUPAC name of the compound

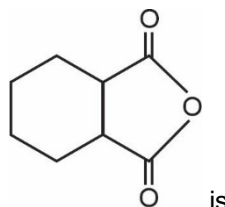


- (1) Methyl propanoate
 (2) Methoxy propanoate
 (3) Methoxy Butanoate
 (4) Methyl Butanoate
3. The correct order of decreasing priority for functional group is
- (1) $-\text{COCl} > -\text{CONH}_2 > -\text{COOR} > \text{SO}_3\text{H}$
 (2) $-\text{SO}_3\text{H} > -\text{COOR} > -\text{COCl} > -\text{CONH}_2$
 (3) $-\text{SO}_3\text{H} > -\text{COCl} > -\text{CONH}_2 > -\text{COOR}$
 (4) $-\text{COOR} > -\text{COCl} > -\text{CONH}_2 > -\text{SO}_3\text{H}$
4. The IUPAC name of the compound



- (1) 3-(Methoxycarbonyl)cyclohexane carboxylic acid
 (2) 3-carboxycyclohexane ethanoate
 (3) 3-methylcarbonylcyclohexane carboxylic acid
 (4) 3-ethanoylcyclohexane carboxylic acid

5. The IUPAC nomenclature of



- (1) Cyclopentane-1, 2-dicarboxylic anhydride
 (2) Hexane-1, 6-dicarboxylic anhydride
 (3) Cyclohexane-1, 2-dicarboxylic acid
 (4) Cyclohexane-1, 2-dicarboxylic anhydride

ARBTS-14

Q.60. (Compounds of Boron)

1. The strongest lewis acid among the following is
- (1) BF_3 (2) BCl_3
 (3) BBr_3 (4) BI_3
2. The number of B—O—B bonds in borax is
- (1) 2 (2) 3
 (3) 4 (4) 5
3. Which of the following is an incorrect statement about Diborane?
- (1) It has two $3c-2e^-$ bond
 (2) It has four $2c-2e^-$ bond
 (3) It cannot undergo hydrolysis
 (4) It reacts with excess of ammonia to give $\text{B}_2\text{H}_6 \cdot 2\text{NH}_3$
4. Consider the reaction
- $$\text{Borax} \xrightarrow{\Delta} \text{A} \xrightarrow{\Delta} \text{B} + \text{C}$$
- B and C are
- (1) $\text{Na}_2\text{B}_4\text{O}_7$, NaBO_2 (2) H_3BO_3 , B_2O_3
 (3) NaBO_2 , B_2O_3 (4) B_2O_3 , $\text{Na}_2\text{B}_4\text{O}_7$
5. Identify the correct statement(s) about orthoboric acid.
- (1) Monobasic lewis acid
 (2) On heating to 100°C it gives metaboric acid
 (3) Layered structure
 (4) All of these

ARBTS-14

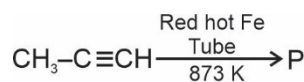
Q.68. (Raoult's Law)

- Which of the following shows positive deviation from Raoult's Law?
 - $\text{HCl} + \text{H}_2\text{O}$
 - $\text{CHCl}_3 + \text{C}_6\text{H}_6$
 - $\text{HNO}_3 + \text{H}_2\text{O}$
 - $\text{C}_2\text{H}_5\text{OH} + \text{H}_2\text{O}$
- Two liquids A and B form an ideal solution at temperature T. The mole fraction of A in the vapour phase and liquid phase are 0.2 and 0.6 respectively. If the total vapour pressure above the solution is 1000 mmHg then the vapour pressure of pure B at temperature T is
 - 1000 mmHg
 - 2000 mmHg
 - 1600 mmHg
 - 2400 mmHg
- A liquid mixture contains 1 mol of P and 1 mol of Q. If the pressure is reduced, at what pressure does the last drop of liquid disappear? ($P_A^0 = 200 \text{ mmHg}$, $P_B^0 = 100 \text{ mmHg}$)
 - 110.98 mmHg
 - 200 mmHg
 - 190 mmHg
 - 133.3 mmHg
- A liquid mixture of benzene and toluene contains 1 mole each. If the pressure over the mixture at TK is reduced, at what pressure does the first drop of vapour form. ($P_{\text{Benzene}}^0 = 103 \text{ mmHg}$ and $P_{\text{Toluene}}^0 = 32.05 \text{ mmHg}$)
 - 87.28 mmHg
 - 59.28 mmHg
 - 67.52 mmHg
 - 72.42 mmHg
- Total vapour pressure of the solution prepared by mixing 2 mol of A and 4 mol of B at 298 K is ($P_A^0 = 200 \text{ atm}$, $P_B^0 = 80 \text{ atm}$)
 - 220 atm
 - 180 atm
 - 120 atm
 - 100 atm

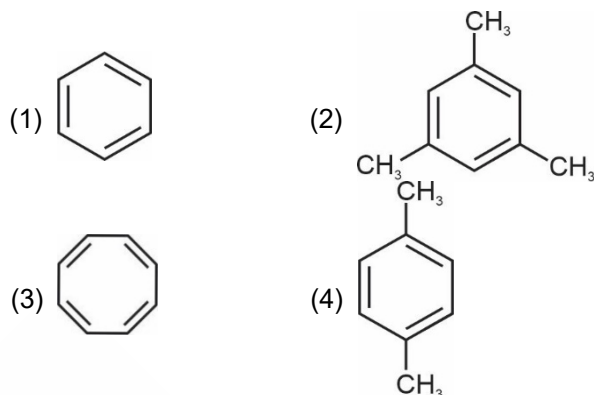
ARBTS-14

Q.65. (Reaction of Alkynes)

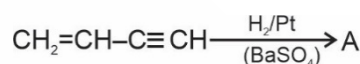
- Consider the reaction



P is

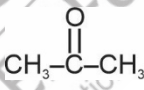
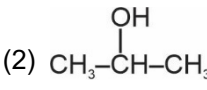
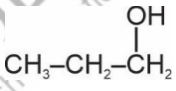


- In the reaction



Identify A (Major)

- $\text{CH}_2=\text{CH--CH=CH}_2$
 - $\text{CH}_3\text{--CH}_2\text{--C}\equiv\text{CH}$
 - $\text{CH}_3\text{--CH}_2\text{--CH}_2\text{--CH}_3$
 - $\text{CH}_2=\text{CH--CH}_2\text{--CH}_3$
- In the reaction

$$\text{CH}_3\text{--C}\equiv\text{CH} \xrightarrow[\text{H}_2\text{O}_2, \text{OH}^-]{\text{BH}_3, \text{THF}} \text{P}$$
 P is
 - 
 - 
 - 
 - $\text{CH}_3\text{--CH}_2\text{--CHO}$
 - Which of the following will form white precipitate with ammonical silver nitrate solution?
 - $\text{CH}_3\text{--C}\equiv\text{CH}$
 - $\text{CH}_3\text{--CH=CH}_2$
 - $\text{CH}_3\text{--C}\equiv\text{C--CH}_3$
 - $\text{CH}_3\text{--CH}_2\text{--CH}_3$
 - On heating But-1-yne with alcoholic KOH the product formed is
 - Butene
 - But-2-ene
 - But-2-yne
 - Butane

CF+OYM
& RMBased on
ARBTS-13 to 15

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CONCEPT STRENGTHENING SHEET

CSS-05 (Chemistry)

Answers Key

Q.88. (Cannizzaro Reaction)

1. (3)
2. (3)
3. (3)
4. (4)
5. (3)

Q.51. (Periodic Table)

1. (3)
2. (1)
3. (1)
4. (4)
5. (4)

Q.82. (Diazotisation)

1. (3)
2. (3)
3. (2)
4. (1)
5. (1)

Q.80. (Optical isomerism)

1. (1)
2. (2)
3. (1)
4. (2)
5. (2)

Q.81. (Decarboxylation Reaction)

1. (1)
2. (2)
3. (2)
4. (3)
5. (2)

Q.82. (Reaction of Ethers)

1. (2)
2. (4)
3. (1)
4. (1)
5. (1)

Q.57. (Nomenclature)

1. (3)
2. (4)
3. (2)
4. (1)
5. (4)

Q.60. (Compounds of Boron)

1. (4)
2. (4)
3. (3)
4. (3)
5. (4)

Q.68. (Raoult's Law)

1. (4)
2. (2)
3. (4)
4. (3)
5. (3)

Q.65. (Reaction of Alkynes)

1. (2)
2. (1)
3. (4)
4. (1)
5. (3)